Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

KE=hc/Î», h=4.14ÃĹ10^-15 eVâĭħns (h=câģĦ2ÏĢ), c=3ÃĹ10^10 cm/sec

Î»=hc/KE

=4.14ÃĹ10^-15 eVâĭħnsÃĹ3ÃĹ10^10 cm/sec/1 eV

=1.242ÃĹ10^-4 cmâīĪ1.23ÃĹ10^-7 cm

</think><answer>G.1.23ÃĹ10^-7 cm</answer>